Author: What type of sequence is impossible then?

San: ① Bounded, monotonic, nonconvergent sequence ② Monotonic, Convergent, Non-bounded sequence ③ Convergent, not monotonic, non-bounded sequence.

Author: In what follows, we shall discuss only convergent sequences. We shall prove the following:

Theorem: $\boxed{\text{A convergent sequence has only one limit.}}$

San: (My proof attempt) Suppose, for the sake of contradiction, let a convergent sequence (y_n) have two limits a_1 and a_2.

- By definition of the limit of sequence, $\forall \varepsilon > 0, \varepsilon \in \mathbb{R}, \exists N \in$ Nat no. s.t. $\forall n \in$ Nat no, $n > N$, $\boxed{|y_n - a_1| < \varepsilon \text{ and } |y_n - a_2| < \varepsilon}$

$\therefore a_1 - \varepsilon < y_n < a_1 + \varepsilon$ and $a_2 - \varepsilon < y_n < a_2 + \varepsilon$

Wlog, let $a_1 > a_2$. $a_1 - \varepsilon > a_2 - \varepsilon$ and $a_1 + \varepsilon > a_2 + \varepsilon$.

$\therefore \forall n > N$, $\boxed{a_2 - \varepsilon < a_1 - \varepsilon < y_n < a_2 + \varepsilon < a_1 + \varepsilon}$

$\boxed{a_1 - a_2 < 2\varepsilon} \rightarrow$ This holds for $\forall \varepsilon > 0$.

$\boxed{a_2 - a_1 < 2\varepsilon} \rightarrow$ This holds for $\forall \varepsilon > 0$

Since both these inequalities must hold at once, $\therefore a_1 > a_2$ and $a_1 < a_2$ will both be false.

$\boxed{\therefore a_1 = a_2}$ holds **Note:** I don't know whether this proof is valid

Author: This is the theorem of the $\boxed{\text{uniqueness}}$ of the limit. It means that a convergent sequence cannot have two or more limits.

Suppose the situation is contrary to the above statement. Consider a convergent sequence with two limits a_1 and a_2 and select a value for $\varepsilon < \frac{|a_1 - a_2|}{2}$. Now assume, for example, that $\varepsilon = \frac{|a_1 - a_2|}{3}$. Since a_1 is a limit, then for the selected value of ε there is N_1 such that for all $n > N_1$ the terms of the sequence (its infinite "tail") must fall inside the interval 1 (See Fig). It means that we must have $|y_n - a_1| < \varepsilon$.

- On the other hand, since a_2 is a limit there is N_2 such that for all $n > N_2$ the terms of the sequence (again its "infinite tail") must fall inside the interval 2.